

scan under Disk and take note of your disk's health status. After running additional diagnostic tools to rule out any hardware issues, the next step is to reset or reinstall your operating system. Windows 10 and Windows 11 offer the option to keep your files, but it's always best to prepare a backup as a precaution. Keep scrolling for more information on how to create backups.

2. Corrupted Data and Accumulating Bad Sectors

If your hard drive is gradually failing, corrupted data can manifest in numerous ways, such as:

- File or folder names that appear scrambled
- Random error messages when opening, moving, or saving files
- Inaccessible files that fail to open
- Corrupted data within your files
- Files or folders that seem to have disappeared

Data corruption can occur during data creation or storage, and it could be caused by a virus or bad sectors on the hard drive. Bad sectors are areas on the hard drive that are unable to maintain data integrity. Usually, Windows automatically hides bad sectors, so they remain unnoticed until data corruption issues arise. However, bad sectors can accumulate quickly on a failing hard drive, leading to more frequent instances of data corruption.

To recover data from bad sectors and prevent future use, you can use the Windows CHKDSK command tool. Follow these steps for a quick scan:

- Press Windows + E to open File Explorer and go to This PC.
- Right-click the failing disk or partition, and select Properties.
- In Properties, switch to the Tools tab and click on Check.
- If Windows indicates that "You don't need to scan this drive," you can still click Scan drive to run the tool. After the scan is complete, you can choose to fix any errors found.

To conduct a comprehensive CHKDSK scan, which can take an extended period and involves a system reboot, follow these steps:

- Right-click Start and select Command Prompt (Admin) to open an Administrator Command Prompt.
- Enter the following command to recover data and fix errors: `chkdsk /r c:` (replace c: with the drive letter you want to scan).

Press Y when prompted, and CHKDSK will start running once you restart your computer.

This process can take several hours, and it's best to run it overnight when you can spare your computer. CHKDSK will scan the entire drive for bad sectors, attempt to recover any data, and mark any bad sectors so they are not used in the future.

3. Strange Sounds

If you start to hear odd sounds emanating from your hard drive, it's a cause for concern. The click of death, a repeating sound that occurs when the head attempts to write data, fails, and repeats the process, is one such noise. Grinding or screeching noises indicate that specific hardware components, such as the bearings or spindle motor, are deteriorating.

Maintenance Tips for your HDD

Here's how you can prevent issues with your hard drive:

1. Monitor Software Downloads

When downloading and installing software on your computer, it's important to exercise caution and only download from trusted sources. Additionally, try to avoid downloading large applications frequently as this can slow down your hard drive. It's crucial to be aware of the risk of viruses infecting your computer and potentially causing damage to the entire system.

To prevent virus-related problems, it's essential to install an anti-virus program and ensure that it's always up-to-date and running.

2. Update your OS

Regularly updating your computer's operating system is essential to ensure the device runs smoothly and to help prevent potential hard drive failures. Don't delay when you receive a notification to update.

3. Get rid of unwanted programs

It's a good practice to regularly review the programs installed on your computer's hard drive. Are there any programs that you no longer use? It's common to have programs that have been left unused for some time. You should perform regular checkups to identify these programs.

4. Defragment your hard drive

The hard drive is a crucial component that works tirelessly to enable the computer to perform its functions. When you install a program or save a file, it uses the first available empty space on the drive. However, not all the available empty spaces are utilized fully, which makes it harder for the computer to access stored data or files.

To maintain the hard drive's good condition, defragmentation is necessary. It is recommended to defragment the hard drive at least once a week to ensure it functions optimally.

5. Clean your hard drive

It is important to not only clean the hard drive through software but also physically clean it. Regularly dusting off the hard drive can help remove any debris that may have accumulated. Physical cleaning of the hard drive is crucial for maintaining proper airflow, preventing overheating, and ensuring other computer components function properly.

6. Protect from power surges

A power surge is a common cause of hard drive failure. It happens when the voltage spikes above normal levels for a few nanoseconds, damaging any electronic device, including your computer. Even a single power surge can fry your hard drive. To avoid such damage, it is recommended to use an uninterrupted